

Operation Research

- ① Linear programming $\rightarrow$ graphical, simplex, Big M, two phase, ^{dual}
- ② Transportation $\rightarrow$ NE, Min cost, Morse, rest of op, ^{dual}
- ③ Assignment $\rightarrow$
- ④ Dynamic
- ⑤ Game
- ⑥ Queuing Th.
- ⑦ Pert / CPM
- ⑧ Inventory
- ⑨ Simulation
- ⑩ Sequencing

Manav Raina

Characteristics, Scope, Methodologies, Modeling

History of OR:

British Army (WWII) $\rightarrow$ US Army LPP

↓
Group of scientists

↓
Scientific Approach

70's, 80 $\rightarrow$ Agriculture, Business ^{houses} ~~class~~

Defination of OR :-

limited resources ^{scientific way} Man made system best design of operation

$\rightarrow$ op of a system $\rightarrow$ scientific tool of technique
(with optimum soln)

$\rightarrow$ Research in operation

$\rightarrow$ Art of winning war without fighting.

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optimization of objective fn by either maximisation of profit or minimisation of cost with considering resource constraints.

Features / Characteristics

- use of Interdisciplinary Teams
- complex system orientation
- Scientific Method
- Quality in decisions

Scope of OR

Planning, Agriculture, cooperates, transportation, Military

Limitation of OR

- Lack of Qualitative factors
- limited factor consideration
- sp category of problems
- resistance of employees.

Manner
Ruina

Scope

① Agriculture :- The explosion of population & consequent shortage of food, every country is facing the problem of -

① optimum allocation of land to various crops in accordance with the climatic condns.

② optimum distribution of water from various res like canal for irrigation purpose.

Thus, there is a need of determining best policies under prescribed restriction. Hence good amount of work can be done

② Marketing :- with the help of OR technique a marketing Administrator can decide

i) distribute the product for sale so that the total cost of transportation etc is min.

ii) Min per unit sale price

iii) Size of the stock to meet future demand

iv) How to select the best advertising media w.r.t time, cost, etc.

③ Personal Management : - A personal manager can use OR techniques:

- i) To appoint the most suitable persons on min salary.
- ii) To determine the best age of retirement for employees
- iii) To find out the no of persons to be appointed on full time basis when the workload is seasonal.

④ LIC : - OR approach is also applicable to enables LIC offices to decide:

- i) what should be the premium rate for various modes of policies.
- ii) how best the profit could be distributed in the cases of with profit policies, etc.

Modelling in OR

- ① classification by structure
- ② classification by purpose
- ③ classification by Nature of Environment
- ④ classification by behaviour
- ⑤ classification by Method of solution

Linear Programming Problems (LPP)

$$\text{Max } Z = x + 2y$$

$$\text{Min } Z = 2x + 3y$$

In LPP there are 2 things one is objective fun & other one is constraint (i.e. availability or requirement)

Graphical Method to solve LPP

$$\text{Max } Z = 3x_1 + 4x_2$$

$$x_1 + 2x_2 \leq 4$$

$$3x_1 + 2x_2 \leq 6$$

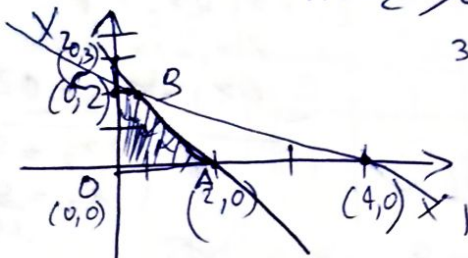
$$x_1, x_2 \geq 0$$

x_1	4	0
x_2	0	2

x_1	2	0
x_2	0	3

less than hoge toh origin ke ander hoge

greater than hoge toh origin ke bahar hoge



Fessible region

$$3(1) + 4(\frac{3}{2}) = 3 + \frac{12}{2} = \frac{6+12}{2} = \frac{18}{2} = 9$$

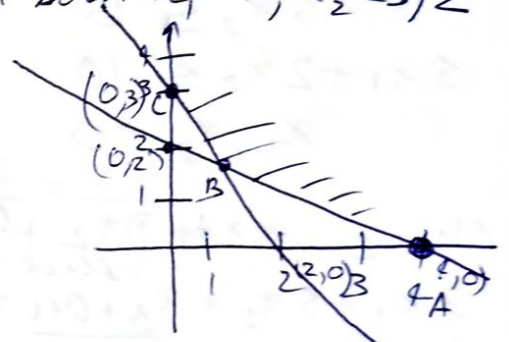
O (0,0)	Z at O = 0
A (2,0)	Z at A = 6
B (1, 3/2)	Z at B = 9
C (0,2)	Z at C = 8

Z is max at B point (1, 3/2) i.e. optimal soln $x_1 = 1, x_2 = 3/2$

$$\text{Max } Z = 9$$

$$\text{Min } Z = 3x_1 + 4x_2$$

A (4,0)	Z at A = 12
B (1, 3/2)	Z at B = 9
C (0,3)	Z at C = 12



Z is min at B point (1, 3/2) i.e. optimal soln $x_1 = 1, x_2 = 3/2$

$$\text{Min } Z = 9$$

Unbounded solution are those soln in which there are more point in feasible soln. No solution

Infinite soln

$$\text{Max } Z = 3x_1 + 2x_2$$

$$x_1 + 2x_2 \leq 4$$

$$3x_1 + 2x_2 \leq 6$$

$$x_1, x_2 \geq 0$$

$$O (0,0), Z at O = 0$$

$$A (2,0), Z at A = 6$$

$$B (1, 3/2), Z at B = 6$$

$$C (0,2), Z at C = 4$$

Z is Max at A & B point so

there is infite soln.

$$\text{Max } Z = 6x_1 - 4x_2$$

$$2x_1 + 4x_2 \leq 4$$

$$4x_1 + 8x_2 \geq 16$$

$$x_1, x_2 \geq 0$$



~~Max~~ $Z = 2x_1 + 3x_2$

Min

$x_1 + x_2 \leq 4$ (1)

$6x_1 + 2x_2 \geq 8$ (2)

$x_1 + 5x_2 \geq 4$ (3)

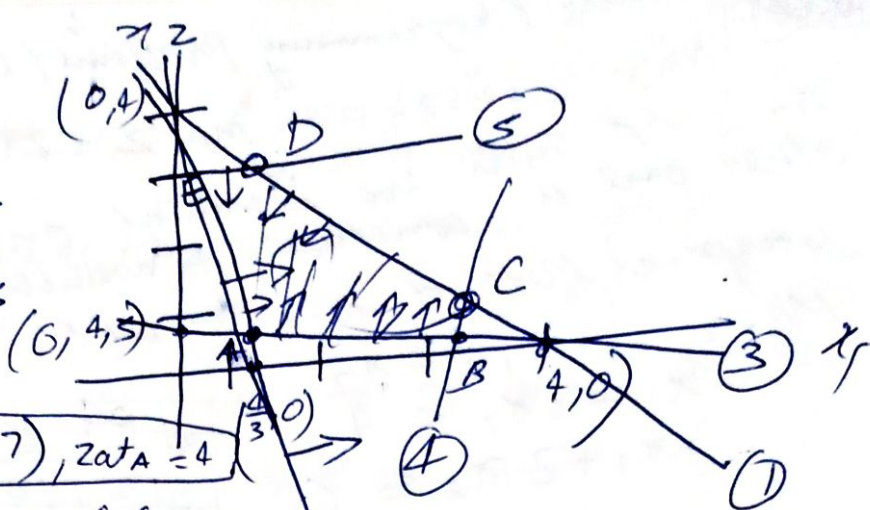
$x_1 \leq 3, x_2 \leq 3$ (4) (5)

$x_1 \geq 0, x_2 \geq 0$

x_1	4	0
x_2	0	4

x_1	8/6	0
x_2	0	4

x_1	4	0
x_2	0	4/5



By (1) & (3) A (1.14, 0.57), $z_{atA} = 4$

By (3) & (4) B (3, 1.5), $z_{atB} = 6.6$

By (1) & (4) C (3, 1), $z_{atC} = 9$

By (1) & (5) D (1, 3), $z_{atD} = 11$

By (2) & (5) E (1/3, 3), $z_{atE} = 9.66$

$$\begin{array}{rcl} 2 \quad 6x_1 + 6x_2 & = & 24 \\ 6x_1 + 2x_2 & = & 8 \\ \hline 6x_1 + 4x_2 & = & 24 \\ -12x_2 & = & 81 \\ x_2 & = & -8.25 \end{array}$$

$$\begin{array}{r} 32 \\ -24 \\ \hline 8 \end{array}$$

Z is Min at $x_1 = 1.14$ & $x_2 = 0.57$ & Min $Z = 4$

Simplex Method

Max $Z = 5x_1 + 3x_2$

$3x_1 + 5x_2 \leq 15$

$5x_1 + 2x_2 \leq 10$

$x_1, x_2 \geq 0$

$C_j - Z_j$ Most positive
Most negative

$-4x_2 \geq 0$
 $x_2 = 0$
 $3x_1 + 5x_2 = 15$
 $5x_1 + 2x_2 = 10$
 $x_1 = 2$
 $x_2 = 1$

Max $Z = 5x_1 + 3x_2 + 0x_3 + 0x_4$

$3x_1 + 5x_2 + x_3 + 0x_4 = 15$ (slack variable)

$5x_1 + 2x_2 + 0x_3 + x_4 = 10$

$x_1, x_2, x_3, x_4 \geq 0$

CB	x_B	C_j b	5 x_1	3 x_2	0 x_3	0 x_4	Min Ratio
0	x_3	15	3	5	1	0	$15/3 = 5$
0	x_4	10	5	2	0	1	$10/5 = 2 \leftarrow$

leaving variable

Z_j	0	0	0	0
C_j	5	3	0	0
$C_j - Z_j$	+5	+3	0	0

entering variable

$$\begin{array}{r} 5 \times 1 \\ 5 \times \frac{1}{5} \\ 2 \times \frac{1}{5} \end{array}$$

C_B	x_B	C_j	S	3	0	0
		B	x_1	x_2	x_3	x_4
0	x_3	9	0	$19/5$	1	$-3/5$
5	x_1	2	1	$2/5$	0	$1/5$
	Z_j		5	2	0	1
	C_j		5	3	0	0
	$C_j - Z_j$		0	1	0	-1

Min Ratio

$$\begin{aligned}
 0L - 3NL \\
 15 - 3(2) &= 9 \\
 3 - 3 &= 0 \\
 5 - 3(2/5) &= 10/5 \\
 1 - 3(0) &= 1 \\
 0 - 3(1/5) &= -3/5
 \end{aligned}$$

$$\begin{aligned}
 9/19/5 &= 45/19 \\
 10/2 &= 5
 \end{aligned}$$

C_B	x_B	C_j	S	3	0	0
		B	x_1	x_2	x_3	x_4
3	x_2	$45/19$	0	1	$5/19$	$-3/19$
5	x_1	$20/19$	1	0	$-2/19$	$5/19$

$$Z_j \quad 5 \quad 3 \quad 5/19 \quad 16/19$$

$$C_j \quad 5 \quad 3 \quad 0 \quad 0$$

$$C_j - Z_j \quad 0 \quad 0 \quad -5/19 \quad -16/19$$

$$x_1 = 20/19, \quad x_2 = 45/19$$

Max

$$5x_1 + 3x_2$$

$$5\left(\frac{20}{19}\right) + 3\left(\frac{45}{19}\right) = \frac{100}{19} + \frac{135}{19} = \frac{235}{19}$$

Q)

$$\text{Max } Z = 3x_1 + 2x_2 + 5x_3$$

$$x_1 + x_2 + x_3 \leq 9$$

$$2x_1 + 3x_2 + 5x_3 \leq 30$$

$$2x_1 - x_2 - x_3 \leq 8$$

$$x_1, x_2, x_3 \geq 0$$

$$\text{Max } Z = 3x_1 + 2x_2 + 5x_3 + 0x_4 + 0x_5 + 0x_6$$

$$x_1 + x_2 + x_3 + x_4 + 0x_5 + 0x_6 = 9$$

$$2x_1 + 3x_2 + 5x_3 + 0x_4 + x_5 = 30$$

$$2x_1 - x_2 - x_3 + 0x_4 + 0x_5 + x_6 = 8$$

$$x_1, x_2, x_3, x_4, x_5, x_6 \geq 0$$

$$\begin{aligned}
 0L - 2/5NL \\
 2 - 2/5(45/19) &= 10/19 \\
 1 - 2/5(0) &= 1 \\
 2/5 - 2/5(1) &= 0 \\
 0 - 2/5(5/19) &= -2/19 \\
 1/5 - 2/5(-3/19) &= 11/19
 \end{aligned}$$

$$3\left(\frac{5}{19}\right) - 5\left(\frac{2}{19}\right)$$

$$\frac{15}{19} - \frac{10}{19} = \frac{5}{19}$$

$$3\left(-\frac{3}{19}\right) + 5\left(\frac{5}{19}\right)$$

$$-\frac{9}{19} + \frac{25}{19} = \frac{16}{19}$$

C_B	x_B	B	x_1	x_2	x_3	x_4	x_5	x_6	Min Ratio
0	x_4	9	1	1	1	1	0	0	$9/1 = 9$
0	x_5	30	2	3	5	0	1	0	$30/5 = 6 \leftarrow$
0	x_6	8	2	-1	-1	0	0	1	$8/(-1)$ div indend cannot do

Z_j 0 0 0 0 0 0
 C_j 3 2 5 0 0 0
 $C_j - Z_j$ 3 2 5 0 0 0

$OL + NL$
 $8 + 6$
 $2 + 2/5$
 $-1 + 3/5$
 $-1 + 1$
 $0 + 0$
 $0 + 1/5$
 $1 + 0$

C_B	x_B	B	x_1	x_2	x_3	x_4	x_5	x_6
0	x_4	3	$3/5$	$2/5$	0	1	$-1/5$	0
5	x_3	6	$2/5$	$3/5$	1	0	$1/5$	0
0	x_6	14	$12/5$	$-2/5$	0	0	$1/5$	1

$Min Ratio$
 $3/3 = 1$
 $6/2 = 3$
 $14/12 = 1.16$
 $OL - NL$
 $= 5 - 9 = -4$
 $= 15 - 1 = 14$
 $= 1 - 1 = 0$
 $1 - 0 = 1$
 $0 - 1/5 = -1/5$
 $0 - 0 = 0$

Z_j 2 3 5 0 1 0
 C_j 3 2 5 0 0 0
 $C_j - Z_j$ +1 -1 0 0 -1 0

$1 - \frac{3}{5}$
 $\frac{5-3}{5} = \frac{2}{5}$

C_B	x_B	B	x_1	x_2	x_3	x_4	x_5	x_6	Min Ratio
3	x_1	5	1	$2/3$	0	$5/3$	$-1/3$	0	
5	x_3	4	0	$1/3$	1	$-2/3$	$1/3$	0	
0	x_6	2	0	$-2/3$	0	-4	1	1	

Z_j 3 ~~$11/3$~~ 5 $5/3$ $-2/3$ 0
 $C_j - Z_j$ 0 ~~$-5/3$~~ 0 $-5/3$ $2/3$ 0

$3(\frac{2}{3}) + 5(\frac{1}{3}) = 11/3$
 $\frac{6}{3} + \frac{5}{3} = 11/3$
 $3(\frac{5}{3}) + 5(\frac{-2}{3}) = 15/3 - 10/3 = 5/3$
 $3(-1/3) + 5(1/3) = -1 + 5/3 = 2/3$
 $3(-2/3) + 5(1/3) = -2 + 5/3 = -1/3$

$\frac{1}{7} - \frac{1}{7}(1)$

$2 - \frac{11}{3} = -\frac{5}{3}$
 $\frac{6-11}{3} = -\frac{5}{3}$
 $0 - \frac{5}{3} = -\frac{5}{3}$

$Max = 3(5) + 2(0) + 4(4) = 15 + 20 = 35$
 $x_1 = 5$
 $x_2 = 0$
 $x_3 = 4$

Two Phase Method

(Big M - Method)

$$\text{Min } z = x_1 + x_2$$

Assume $M=100$

$$2x_1 + x_2 \geq 4$$

$$x_1 + 7x_2 \geq 7$$

$$x_1, x_2 \geq 0$$

$$\text{Max } z' = -x_1 - x_2 + 0x_3 + 0x_4 - Mx_5 - Mx_6 \quad - (1+3000)$$

$$2x_1 + x_2 - x_3 + 0x_4 + x_5 + 0x_6 = 4 \quad -1 \quad -3000$$

$$x_1 + 7x_2 + 0x_3 - x_4 + 0x_5 + x_6 = 7 \quad -1 \quad -8000$$

surplus

$$x_1, x_2, x_3, x_4, x_5, x_6 \geq 0$$

C _B	x _B	b	x ₁	x ₂	x ₃	x ₄	x ₅	x ₆	
-M	x ₅	4	2	1	-1	0	1	0	4/1 = 4
-M	x ₆	7	1	7	0	-1	0	1	7/7 = 1 leaving
Z _j			-3M	-8M	M	M	-M	-M	
C _j - Z _j			-1-3M	-1-8M	-M	-M	0	0	

C _B	x _B	b	x ₁	x ₂	x ₃	x ₄	x ₅	x ₆	Min Ratio
-M	x ₅	3	13/7	0	-1	1/7	1	-1/7	21/13 leaving
-1	x ₂	1	1/7	1	0	-1/7	0	1/7	7
Z _j			-13M-1	-1	+M	M/7+1/7	-M	M+1	

$$C_j - Z_j = -1(-13M-1) = 13M+1$$

↑ entering

13 x 7 = 91

7 13

$$\begin{aligned} 0L-NL & 4-1=3 \\ -6M-1/7 & 2-1/7=13/7 \\ 2-1/7 & 1-1=0 \\ 14-1 & -1-0=-1 \\ 7 & 0-1/7= \\ -7M-M & 1-0=1 \\ -6M-1 & 0-1/7 \end{aligned}$$

C_B	x_B	b	x_1	x_2	x_3	x_4	
-1	x_1	21/13	1	0	-7/13	+1/13	$\frac{7-7 \times 13}{13} = \frac{49-91}{13} = \frac{-44}{13}$
-1	x_2	10/13	0	1	7/91	-1/91	$\frac{1+12}{13 \times 91} = \frac{13}{1183}$
	$z_j - c_j$		-1	-1	-42/91	19/91	$\frac{7+12}{91} = \frac{19}{91}$

$C_j - 2^0$
 $-1 + 1$
 -1

$$x_1 = 21/13$$

$$x_2 = 10/13$$

$$\boxed{\text{Min } Z = \frac{21}{13} + \frac{10}{13} = \frac{31}{13}}$$

2 Phase Method

$$\text{Min } Z = x_1 + x_2$$

$$2x_1 + x_2 \geq 4$$

$$x_1 + 7x_2 \geq 7$$

$$x_1, x_2 \geq 0$$

Phase 1

$$\text{Max } Z' = +0x_1 + 0x_2 + 0x_3 + 0x_4 - x_5 - x_6$$

$$2x_1 + x_2 - x_3 + 0x_4 + x_5 + 0x_6 = 4$$

$$x_1 + 7x_2 - x_4 + 0x_5 + 0x_6 = 7$$

C_B	x_B	b	x_1	x_2	x_3	x_4	x_5	x_6	Min Ratio
-1	x_5	4	2	1	0-1	0	1	0	4/1=4
-1	x_6	7	1	7	0	-1	0	1	7/7=1
	$C_j - 2^0$		0	0	1	1	0	0	

$0L - \frac{1}{7}NL$
 $1 - \frac{1}{7}(21/13)$
 $117 - 117(1)$
 $1 - 117(0)$
 $0 - 1/2(-7/13)$
 $-1/7 - 1/7(-1/13)$
 2
 $\frac{13}{91}$
 $1 - \frac{21}{13 \times 7}$
 $\frac{91-21}{91} = \frac{70}{91}$
 8
 $\frac{11}{42}$
 2
 $\frac{13}{91}$
 $-\frac{1}{7}(-\frac{7}{13})$
 $\frac{7}{91}$
 $-\frac{1}{7} - \frac{1}{7}(-\frac{1}{13})$
 $-\frac{1}{7} + \frac{1}{91}$
 $-\frac{13+1}{91} = \frac{-14}{91}$
 $1 - \frac{1}{7}(21/13)$
 $\frac{10}{70}$
 $\frac{70}{91}$
 13

		C_j	0	0	0	0	-1	
C_B	x_B	b	x_1	x_2	x_3	x_4	x_5	Min Ratio
-1	x_5	3	13/2	0	-1	1/2	1	21/13
0	x_2	1	1/2	1	0	-1/2	0	7

$$C_j - Z_j \quad 13/2 \quad 0 \quad -1 \quad 1/2 \quad 0$$

		C_j	0	0	0	0
C_B	x_B	b	x_1	x_2	x_3	x_4
0	x_1	21/13	1	1	-7/13	1/13
0	x_2	10/13	0	0	1/13	-2/13

Phase 2 $\text{Max } Z' = -x_1 - x_2 + 0x_3 + 0x_4$

C_B	x_B	b	-1	-1	0	0
-1	x_1	21/13	1	1	-7/13	1/13
-1	x_2	10/13	0	0	1/13	-2/13
$C_j - Z_j$			0	0	-6/13	-1/13

$$-1\left(-\frac{7}{13}\right) - 1\left(\frac{1}{13}\right) + \frac{7}{13} - \frac{1}{13} = \frac{6}{13}$$

$$x_1 = 21/13$$

$$x_2 = 10/13$$

$$\frac{21}{13} + \frac{10}{13} = \frac{31}{13}$$

Duality in LPP

$$\text{Max } Z_P = 3x_1 + 5x_2$$

$$2x_1 + 3x_2 \leq 8$$

$$x_1 + 2x_2 \leq 5$$

$$x_1, x_2 \geq 0$$

$$\text{Min } Z_D = 8w_1 + 5w_2$$

$$2w_1 + w_2 \geq 3$$

$$3w_1 + 2w_2 \geq 5$$

$$w_1, w_2 \geq 0$$

$$\text{Max } Z_P = 3x_1 + x_2$$

$$x_1 + 4x_2 \leq 7$$

$$3x_1 + x_2 \leq 9$$

$$x_1, x_2 \geq 0$$

$$\text{Min } Z_D = 7w_1 + 9w_2$$

$$w_1 + 3w_2 \geq 3$$

$$4w_1 + w_2 \geq 1$$

$$w_1, w_2 \geq 0$$

$$\textcircled{2} \text{ Min } z = x_1 + 4x_2$$

$$3x_1 + x_2 \geq 2$$

$$2x_1 + 5x_2 \geq 9$$

$$x_1, x_2 \geq 0$$

$$\text{Max } 7w_1 + 9w_2$$

$$3w_1 + 2w_2 \leq 1$$

$$2w_1 + 5w_2 \leq 4$$

$$w_1, w_2 \geq 0$$

$$\textcircled{3} \text{ Max } z = 3x_1 - x_2 + x_3$$

$$4x_1 - x_2 \leq 8$$

$$8x_1 + x_2 + 3x_3 \geq 12$$

$$5x_1 - 6x_2 \leq 13$$

$$x_1, x_2, x_3 \geq 0$$

$$\text{Max } z = 3x_1 - x_2 + x_3$$

$$4x_1 - x_2 + 0x_3 \leq 8$$

$$8x_1 - x_2 - 3x_3 \leq 12$$

$$5x_1 + 6x_2 + 0x_3 \leq 13$$

$$x_1, x_2, x_3 \geq 0$$

$$\text{Min } z_D = 8w_1 + 12w_2 + 13w_3$$

$$4w_1 - 8w_2 + 5w_3 \geq 3$$

$$-w_1 - w_2 + 6w_3 \geq 1$$

$$-3w_2 \geq 1$$

$$w_1, w_2, w_3 \geq 0$$

$$2(0) + 2(4) = 8$$

$$2 + 4(2) = 10$$

$$1 + 2 = 3$$

$$10$$

$$\textcircled{4} \text{ Max } z = x_1 + 3x_2$$

$$3x_1 + 2x_2 \leq 6$$

$$3x_1 + x_2 = 4$$

$$x_1, x_2 \geq 0$$

$$\text{Min } z_p = 6w_1 + 4w_2$$

$$3w_1 + 3w_2 \geq 1$$

$$2w_1 + w_2 \geq 3$$

$$w_1 \geq 0, w_2 \text{ (unrestricted)}$$

$$\text{Max } z_p = x_1 + 3x_2$$

$$3x_1 + 2x_2 \leq 6$$

$$3x_1 + x_2 \leq 4$$

$$3x_1 + x_2 \geq 4$$

$$x_1, x_2 \geq 0$$

$$\text{Max } z_p = x_1 + 3x_2$$

$$3x_1 + 2x_2 \leq 6$$

$$3x_1 + x_2 \leq 4$$

$$-3x_1 - x_2 \leq -4$$

$$x_1, x_2 \geq 0$$

$$\begin{array}{c|c|c} x_1 & 0 & 10 \\ \hline x_2 & 3 & 0 \end{array}$$

$$\begin{array}{c|c|c} x_1 & 0 & 1 \\ \hline x_2 & 1 & 0 \end{array}$$

$$\text{Min } z_p = 6w_1 + 4w_2$$

$$A(1,0) = 1$$

$$B(10,0) = 10$$

$$C(4,4) = 10$$

$$D(0,4) = 8$$

$$E(0,1) = 2$$

$$x_1 + 2x_2$$

$$x_1 + 2x_2 \leq 10$$

$$x_1 + x_2 \geq 1$$

$$x_2 \leq 4$$

$$\text{Min } z_p = 6w_1 + 4w_2' - 4w_2''$$

$$3w_1 + 3w_2' + 3w_2'' \geq 1$$

$$2w_1 + w_2' + w_2'' \geq 3$$

$$\text{Min } z_p = 6w_1 + 4(w_2' - w_2'')$$

$$3w_1 + 3(w_2' + w_2'') \geq 1$$

$$2w_1 + (w_2' + w_2'') \geq 3$$

$$\text{let } w_2' - w_2'' = w_2$$

$$\text{Min } z_p = 6w_1 + 4w_2$$

$$3w_1 + 3w_2 \geq 1$$

$$2w_1 + w_2 \geq 3$$

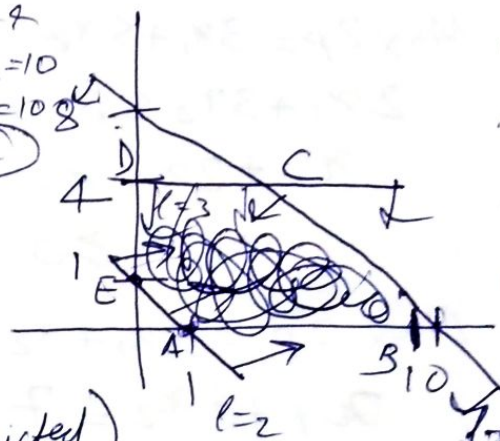
$$w_1 \geq 0, w_2 \text{ (unrestricted)}$$

$$x_2 = 4$$

$$x_1 + 2x_2 = 10$$

$$x_1 + 8 = 10$$

$$x_1 = 2$$



Sensitivity analysis :- It refers to an analysis of the effect on the optimal soln of changes in the parameters of problem on current optimal soln. Sometimes it is also called post-optimality analysis

4 types changes

- ① Change in objective fn
 - Variable included in soln
 - Variable not included in soln
- ② Change in b (right hand side ranging)
- ③ Changes in technology coefficient (a_{ij}'s)
- ④ Adding or deleting constraints.

Formulation

$$TM = 10$$

$$P_{\text{last}} = 30 \text{ days}$$

$$10 \text{ hrs} \geq 7 \text{ complete}$$

$$3 \rightarrow \text{mhi ltr}$$

$$\text{Jan } 100, \text{ Feb } 150, \text{ March } 200, \text{ April } 250, \text{ Total } 130$$

$$\text{Min } z$$

$$400(10x_1 + 10x_3 + 10x_5) + 700(100x_1 + x_3 + x_5) + 500(x_2 + x_4 + x_6)$$

$$x_1 + x_2 = 30$$

$$7x_1 - x_3 - x_4 = 20$$

$$7x_1 + 7x_3 - x_5 - x_6 = 70$$

$$7x_1 + 7x_3 + 7x_5 = 120$$

$$100 + x_1 + x_2 = 130$$

$$150 + x_3 + x_4 = 130 + 7x_1$$

$$200 + x_5 + x_6 = 130 + 7x_1 + 7x_3$$

$$130 + 7x_1 + 7x_3 + 7x_5 = 250$$

$$\begin{array}{r} 250 \\ 130 \\ \hline 120 \end{array}$$

$$4000x_1 + 4000x_3 + 4000x_5 + 700x_1 + 700x_3 + 700x_5 + 500x_2 + 500x_4 + 500x_6$$

$$4700x_1 + 500x_2 + 4700x_3 + 500x_4 + 4700x_5 + 500x_6$$

$$x_1 + x_2 = 30$$

$$-x_3 - x_4 + 7x_1 = 20$$

$$7x_1 + 7x_3 - x_5 - x_6 = 70$$

$$7x_1 + 7x_3 + 7x_5 = 120$$

$$Q \quad z = 3x_1 + 2x_2$$

$$-x_1 + 2x_2 \leq 4$$

$$3x_1 + 2x_2 \leq 14$$

$$x_1 - x_2 \leq 3$$

7 -

$$\text{Max } z = 3x_1 + 2x_2 + 0x_3 + 0x_4 + 0x_5$$

$$-x_1 + 2x_2 + x_3 + 0x_4 + 0x_5 = 4$$

$$3x_1 + 2x_2 + 0x_3 + x_4 + 0x_5 = 14$$

$$x_1 - x_2 + 0x_3 + 0x_4 + x_5 = 3$$

$$\frac{4}{\frac{14}{3}}$$

$$x_1, x_2, x_3, x_4, x_5 \geq 0$$

C_B	x_B	b	x_1	x_2	x_3	x_4	x_5	Min Ratio
0	x_3	4	-1	2	1	0	0	$4/-1$ N.D. ← Not pos
0	x_4	14	3	2	0	1	0	$14/3 = -$
0	x_5	3	1	-1	0	0	1	$3/1 = 3$ ← leaving

$$z_j \quad 0 \quad 0 \quad 0 \quad 0 \quad 0$$

$$C_j - z_j \quad 3 \quad 2 \quad 0 \quad 0 \quad 0$$

$$3(4) + 2(1)$$

$$12 + 2 = 14$$

C_B	x_B	b	x_1	x_2	x_3	x_4	x_5	Min Ratio
0	x_3	7	0	1	1	0	1	$7/1 = 7$
0	x_4	5	0	5	0	1	-3	$5/5 = 1$ ← leaving $3 \rightarrow -2$
3	x_1	3	1	-1	0	0	1	$3/-1 = -$

$$z_j \quad 3 \quad -3 \quad 0 \quad 0 \quad 3$$

$$C_j - z_j \quad 0 \quad -5 \quad 0 \quad 0 \quad -3$$

$$C_j \quad 3 \quad 2 \quad 0 \quad 0 \quad 0$$

$$-1+1$$

$$0L+3NL$$

$$14-3(3) \quad 2+3$$

$$3-3(1)$$

$$2-3(-1) \quad 600$$

$$0-3(0)$$

$$1-3(0)$$

$$0-3(1)$$

C_B	x_B	b	x_1	x_2	x_3	x_4	x_5
0	x_3	6	0	0	0	-1	2/5
2	x_2	1	0	1	0	1	-3/5
3	x_1	4	1	0	0	1	2/5

$$2(-\frac{3}{5}) + 3(\frac{2}{5})$$

$$-\frac{6}{5} + \frac{6}{5}$$

$$5-3$$

$$1-\frac{3}{5}$$

$$3 \quad 2 \quad 0 \quad 3 \quad 0$$

$$-\frac{3}{5} \quad 0 \quad 0 \quad 0 \quad 0$$

$$0L-NL$$

$$7-1=6$$

$$0-0=0$$

$$1-1=0$$

$$1-0=0$$

$$0-1=-1$$

$$1-(-3/5)$$

$$0L+NL$$

$$3+1=4$$

$$1+0=1$$

$$-1+1=0$$

$$0+0=0$$

$$0+1=1$$

$$1+(-3/5)$$

$$4+3=7$$

$$-1+1=0$$

$$2+(-1)=1$$

$$1+0=1$$

$$0+0=0$$

$$0+1=1$$

$$1-\frac{3}{5}$$

$$5-3=2/5$$

$$(9) \quad Z = -3x_1 + x_2 + x_3$$

Min

$$x_1 - 2x_2 + x_3 \leq 11$$

$$-4x_1 + x_2 + 2x_3 \geq 3$$

$$2x_1 - x_3 = -1$$

$$\text{Max } z' = .3x_1 - x_2 - x_3 + 0x_4 + 0x_5 + Mx_6 + Mx_7$$

$$\begin{aligned} x_1 - 2x_2 + x_3 + x_4 + 0x_5 + 0x_6 + 0x_7 &= 11 \\ -4x_1 + x_2 + 2x_3 + 0x_4 + x_5 + 0x_6 + 0x_7 &= 3 \\ 2x_1 - x_3 + 0x_4 + 0x_5 + 0x_6 + x_7 &= -1 \end{aligned}$$

$$x_1 - 2x_2 + x_3 + x_4 + 0x_5 + 0x_6 + 0x_7 = 11$$

$$-4x_1 + x_2 + 2x_3 + 0x_4 - x_5 + x_6 + 0x_7 = 3$$

$$-2x_1 + x_3 + 0x_4 + 0x_5 + 0x_6 + x_7 = -1$$

$$x_1, x_2, x_3, x_4, x_5, x_6, x_7 \geq 0$$

CB	x_B	b	x_1	x_2	x_3	x_4	x_5	x_6	x_7	Min Ratio
0	x_4	11	1	-2	1	1	0	0	0	11/1 ← leaving
M	x_6	3	-4	1	2	0	-1	1	0	3/-4
M	x_7	1	-2	0	1	0	0	0	1	1/-2 ←

$$z_j - 6M \quad M \quad 3M \quad 0 \quad -M \quad M \quad M \quad -6$$

$$-4M - 2M$$

$$C_j - z_j \quad (-3-6M) \quad (1-M) \quad (1-3M) \quad 0 \quad (M) \quad 0 \quad 0$$

$$\begin{aligned} -3-600 &= -603 \\ 1-100 &= -99 \\ 1-300 &= -299 \\ 100 &= 100 \\ 3-600 &= -597 \end{aligned}$$

CB	x_B	b	x_1	x_2	x_3	x_4	x_5	x_6	Min Ratio
-3	x_1	11	1	-2	1	1	0	0	11/-2
M	x_6	47	0	-7	6	4	-1	1	47/-7
M	x_7	23	0	-4	3	2	0	0	23/-4

$$z_j - 3 \quad (-6-11M) \quad (-3+9M) \quad (-3+6M) \quad (-M) \quad (M)$$

$$C_j - z_j \quad 0 \quad (-5-11M) \quad (-2+9M) \quad (3-6M) \quad (M) \quad (0)$$

$$\begin{aligned} 0L + 2NL &= 23 \\ 1 + 2(11) &= 23 \\ -2 + 2(1) &= 0 \\ 0 + 2(-2) &= -4 \\ 1 + 2(1) &= 3 \\ 0 + 2(1) &= 2 \\ 0 + 2(0) &= 0 \\ 0 + 2(0) &= 0 \end{aligned}$$

$$\begin{aligned} 0L + 4NL &= 47 \\ 3 + 4(11) &= 47 \\ -4 + 4(1) &= 0 \\ 2 - 900 &= -898 \\ -5 - 1100 &= -1105 \\ 2 + 4(-2) &= -6 \\ 2 + 4(1) &= 6 \\ 0 + 4(1) &= 4 \\ -1 + 4(0) &= -1 \\ 1 + 4(0) &= 1 \end{aligned}$$

$$1-3$$

$$-3+3$$

$$0.8$$

$$\frac{3}{4}$$

$$Q) Z = -3x_1 + x_2 + x_3$$

$$x_1 - 2x_2 + x_3 \leq 11$$

$$-4x_1 + x_2 + 2x_3 \geq 3$$

$$2x_1 - x_3 = -1$$

$$x_1 = 4, x_2 = 1, x_3 = 9$$

$$-3(4) + 1 + 9$$

$$-12 + 10 = -2$$

$$\text{Min } Z = -3x_1 + x_2 + x_3 + 0x_4 + 0x_5 + Mx_6 + Mx_7$$

$$x_1 - 2x_2 + x_3 + x_4 + 0x_5 + 0x_6 + 0x_7 = 11$$

$$-4x_1 + x_2 + 2x_3 + 0x_4 - x_5 + x_6 + 0x_7 = 3$$

$$-2x_1 + x_3 + 0x_4 + 0x_5 + 0x_6 + x_7 = 1$$

$$x_1, x_2, x_3, x_4, x_5, x_6, x_7 \geq 0$$

$$\frac{3}{2} \times 1.5$$

CB	XB	CB	b	x_1	x_2	x_3	x_4	x_5	x_6	x_7	Min Ratio
		-3	1	1	0	0	M	M			
x_4	Q	11		-2	1	1	0	0	0	0	11/1 = 11
x_6	M	3		-4	1	2	0	-1	1	0	3/2 = 0.5
x_7	M	1		-2	0	1	0	0	0	1	1/1 = 1 \leftarrow \text{leaving}

$$Z^j \quad -6M \quad M \quad 3M \quad 0 \quad -M \quad M \quad M$$

$$C_j - Z_j \quad -3 - 6M \quad 1 - M \quad 1 - 3M \quad 0 \quad M \quad 0 \quad 0$$

$$\begin{aligned} OL - 2NL \\ 3 - 2(1) \\ -4 + 2(-2) \\ 1 - 2(0) \\ 2 - 2(1) = 0 \\ 0 - 2(0) \\ -1 - 2(0) \\ 1 - 2(0) \end{aligned}$$

CB	XB	CB	b	x_1	x_2	x_3	x_4	x_5	x_6	Min Ratio
		-3	1	1	0	0	M	M		
x_4	Q	10		3	-2	0	1	0	0	
x_6	M	1		0	1	0	0	-1	1	10/-2 = -5
x_3	1	1		-2	0	1	0	0	0	1/1 = 1 \leftarrow \text{leaving}

$$Z^j \quad -2M \quad M \quad 1 \quad 0 \quad -M \quad M$$

$$C_j - Z_j \quad -1 \quad 1 - M \quad 0 \quad 0 \quad M \quad 0$$

$$\begin{aligned} OL - NL \\ 1 + \\ -2 + \\ 0 + \\ 1 + \\ 0 + \end{aligned} \quad \begin{aligned} OL - NL \\ 11 - 1 = 10 \\ 1 + 2 = 3 \\ -2 - 0 \\ 1 - 1 = 0 \\ 1 - 0 \\ 0 - 0 \\ 0 - 0 \end{aligned}$$

CB	XB	CB	b	x_1	x_2	x_3	x_4	x_5	Min Ratio
		-3	1	1	0	0	0	0	
0	x_4	12		3	0	0	1	-2	12/3 = 4
1	x_2	1		0	1	0	0	-1	1/0
1	x_3	1		-2	0	1	0	0	1/-2

$$Z^j \quad -2 \quad 1 \quad 1 \quad 0 \quad -1$$

$$C_j - Z_j \quad -1 \quad 0 \quad 0 \quad 0 \quad 1$$

$$-3 + 2 \quad \frac{1}{3} \quad \frac{1}{3} \quad \frac{1}{3}$$

CB	XB	CB	b	x_1	x_2	x_3	x_4
-3	x_1	4		1	0	0	1/3
1	x_2	1		0	1	0	0
1	x_3	9		0	0	1	2/3

$$\begin{aligned} OL + 2NL \\ 1 + 2(4) = 9 \\ -2 + 2(1) = 0 \\ 0 + 2(0) = 0 \\ 1 + 2(0) = 0 \\ 0 + 2(1/3) \end{aligned} \quad \begin{aligned} OL + 2NL \\ 10 + 2(1) = 12 \\ 3 + 2(0) = 3 \\ -2 + 2(1) = 0 \\ 0 + 2(0) = 0 \\ 0 + 2(-1) = -2 \end{aligned}$$

Min
 $z = -3x_1 + x_2 + x_3$

$x_1 - 2x_2 + x_3 \leq 11$

$-4x_1 + x_2 + 2x_3 \geq 3$

$2x_1 - x_3 = -1$

2 surplus

$$\begin{array}{|c|c|} \hline C_j - Z_j \\ \hline C_j - Z_j \\ \hline \end{array}$$

Min $z = -3x_1 + x_2 + x_3 + 0x_4 + 0x_5 + 0x_6 + 0x_7$

$x_1 - 2x_2 + x_3 + x_4 + 0x_5 + 0x_6 + 0x_7 = 11$

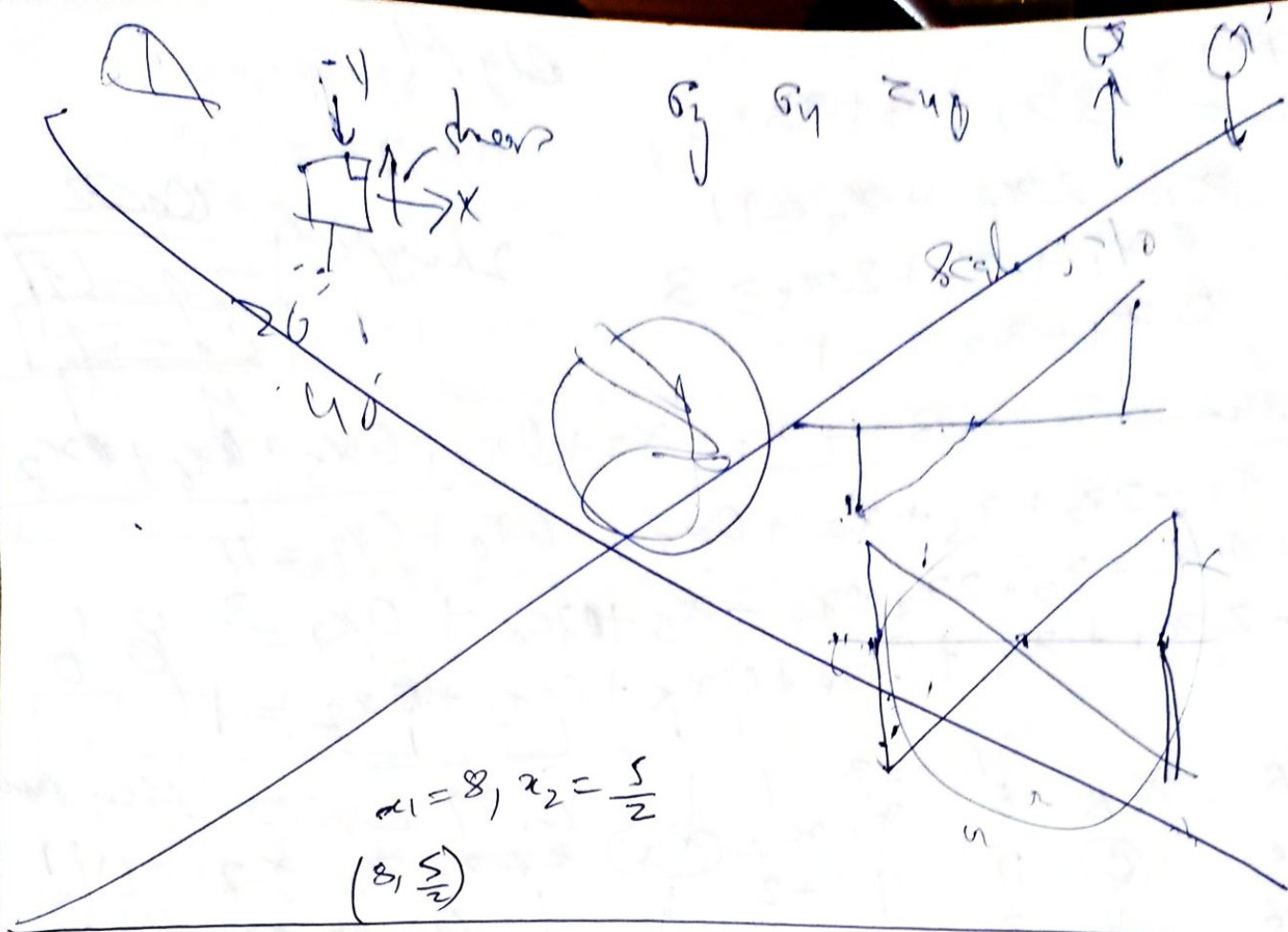
$-4x_1 + x_2 + 2x_3 + 0x_4 - x_5 + 0x_6 + 0x_7 = 3$

$-2x_1 + x_3 + 0x_4 + 0x_5 + 0x_6 + 0x_7 = 1$

C.B	x_B	C_j	-3	1	1	0	M	0	0	Min Ratio
x_4	11		x_1	x_2	x_3	x_4	x_5	x_6	x_7	11/1
x_6	3			-2	1	1	0	0	0	3/2
x_7	1			-4	1	-2	0	-1	1	1/1
	0			-2	0	1	0	0	0	1/1
		Z_j	-6M	M	3M	0	-M	M	M	
		$C_j - Z_j$	-3-6M	1-M	1-3M	0	M	-M	-M	

C.B	x_B	C_j	-3	1	1	0	0	0	0	Min Ratio
x_4	10		x_1	x_2	x_3	x_4	x_5	x_6	x_7	10/1
x_6	1			3	2	0	1	0	0	1/2
x_3	1			0	1	0	0	-1	1	1/1
		Z_j	-2	M	1	0	-M	M	0	
		$C_j - Z_j$	-1	1-M	0	0	M	0	-2M	

C.B	x_B	C_j	-3	1	1	0	0	0	0	Min Ratio
x_4	5		x_1	x_2	x_3	x_4	x_5	x_6	x_7	5/1
x_2	10			0	1	0	0	0	0	10/1
x_3	1			0	1	0	0	0	0	1/1
		Z_j	-4	-2	2	0	-2	2	0	
		$C_j - Z_j$	1	3	-1	0	2	-2	0	



Transportation & Assignment :- LP formulation, transportation algorithm, Hungarian algorithm

Game Theory :- Pure & Mixed strategies, graphical soln, Dominance, LP formulation.

LP formulation

Northwest corner Rule

Least cost Rule

VAM (Vogel Approx Method)

Degeneracy

$$x_2 = 10$$

$$5x_1 + 30 = 45$$

$$5x_1 = 15$$

$$x_1 = 3$$

Standard

Nonstandard

Transp Algorithm

Hungarian Method

$$Z_A = 40(8) + 36(\frac{5}{2}) = 320 + 90 = 410$$

$$Z_B = 320 + 360 = 680$$

$$Z_C = 120 + 360 = 480$$

$$\begin{array}{r} 4 \\ 13 \\ \hline 25 \\ 90 \\ \hline 45 \\ 30 \\ \hline 15 \end{array}$$

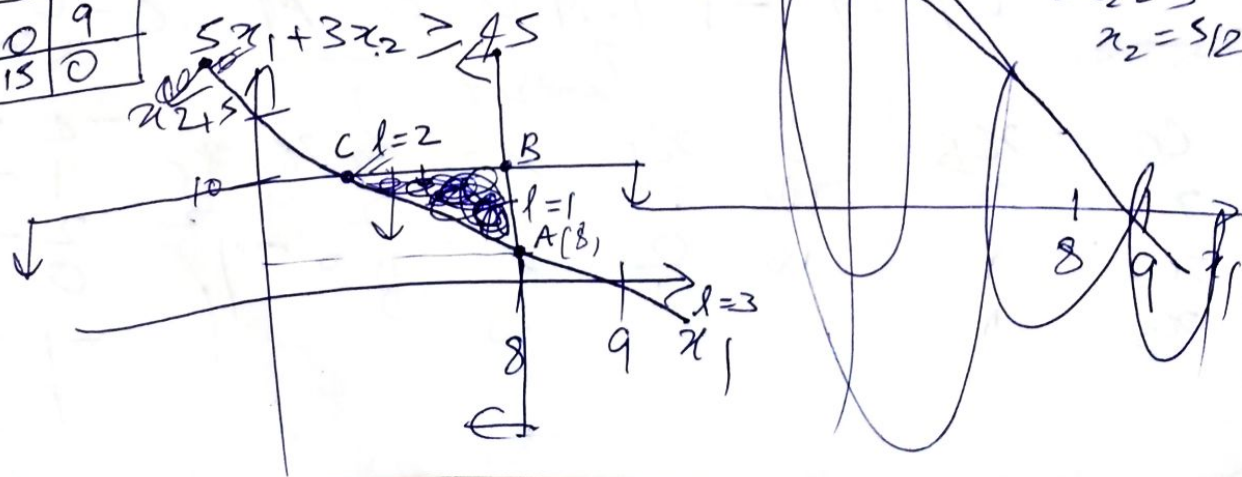
$$z = 40x_1 + 36x_2$$

$$x_1 \leq 8$$

$$x_2 \leq 10$$

$$5x_1 + 3x_2 \geq 45$$

x_1	0	9
x_2	15	0



Transportation in LPP

supply

Origin	Destination				Supply
	1	2	3	...	
1	x_{11} c_{11}	x_{12} c_{12}	x_{13} c_{13}	x_{1n} c_{1n}	a_1
2	x_{21} c_{21}	x_{22} c_{22}	x_{23} c_{23}	x_{2n} c_{2n}	a_2
3	x_{31} c_{31}	x_{32} c_{32}	x_{33} c_{33}	x_{3n} c_{3n}	a_3
...	...	...	...	...	...
m	x_{m1} c_{m1}	x_{m2} c_{m2}	x_{m3} c_{m3}	x_{mn} c_{mn}	a_m
Demand	b_1	b_2	b_3	...	b_n

c_{ij} → cost of transporting one unit of product.

x_{ij} → Quantity transported

Objective :- To transport various quantities that are initially stored at various origins to diff destinations in such a way that the transportation cost is minimum.

Transportation Types

① Unbalanced

② Demand > Supply, $\sum b_i > \sum a_i$

② Balanced

Demand = Supply

③ Demand < Supply, $\sum b_i < \sum a_i$

$\sum a_i = \sum b_i$

Adding a addition row with cost 0 & demand must be eq. to (demand - supply)

③ Demand < Supply, $\sum b_i < \sum a_i$

Adding a addition column with cost 0 & demand = (supply - demand)

	D1	D2	D3	D4	Supply
O1	5	4	2	6	30
O2	8	3	5	7	40
O3	10	11	3	1	20
O4	9	12	6	3	10
D	10	50	30	20	

D = 110
S = 100

	D1	D2	D3	D4	Supply
O1	5	4	2	6	30
O2	8	3	5	7	40
O3	10	11	3	1	20
O4	9	12	6	3	10
O5	0	0	0	0	10
D	10	50	30	20	

$\sum b_i < \sum a_i$

	D1	D2	D3	D4	S
O1	1	9	7	8	50
O2	13	2	3	12	20
O3	15	6	6	11	10
O4	5	10	16	4	30
D	10	20	30	40	

$\sum b_i > \sum a_i$

	D1	D2	D3	D4	D5	S
O1	1	9	7	8	0	50
O2	13	2	3	12	0	20
O3	15	6	6	11	0	10
O4	5	10	16	4	0	30
D	10	20	30	40	10	

D = 100

S = 110

Method of Transportation

① Initial soln :-

- N-W corner rule = $(m+n-1)$
- least cost Method or Matrix Minima Method
- Vogel approximation Method (VAM) - most preferably used

② Optimal soln :-

- MODI Method (Modified Distribution Method)
- Stepping Stone Method.

Balance Demand Transportation (N-W Balanced)

	D1	D2	D3	Supply
01	2	7	4	13
02	3	3	1	7
03	5	4	7	16
04	1	6	2	9
D	11	20	14	34

$7-5 = 2$
 $9-6 = 3$
 $18-4 = 14$
 $D = S$

	D1	D2	D3	S
01	2	7	4	13
02	3	3	1	7
03	5	4	7	16
04	1	6	2	9
D	11	20	14	34

The prob is balanced TP as total supply is equal to total demand, we find the basic feasible soln as given in table

N-W (Unbalanced)

	D1	D2	D3	S
S1	4	8	8	20
S2	16	24	16	56
S3	8	16	24	48
D	28	48	48	124

$D = 124$
 $S = 124$

∴ the min transportation cost is given by

$2 \times 5 + 3 \times 2 + 3 \times 6 + 4 \times 3 + 7 \times 4 + 2 \times 14$
 $10 + 6 + 18 + 12 + 28 + 28 = 102$

The problem is unbalanced TP as the total supply is greater than total demand

$\sum b_j < \sum a_i$

	D1	D2	D3	D4	S
S1	4	8	8	10	30
S2	16	24	16	0	56
S3	8	16	24	6	54
D	28	48	48	16	140

$102-4 = 98$
 $98-82 = 16$

~~D1~~ ~~D2~~ ~~D3~~ ~~Dummy S~~
Initial feasible soln

	D1	D2	D3	D4	S
S1					
S2					
S3					
D					

$$= 4 \times 12 + 8 \times 4 + 24 \times 82 + 16 \times 16 + 24 \times 41 + 0 \times 20$$

$$= ₹ 3528$$

Here no of allocated cell = 6 (m+n-1).
So the soln is non-degenerate.

Initial feasible soln

Least cost Method (Balanced)

	D1	D2	D3	D4	S
O1	6	4	1	5	14
O2	8	9	2	7	16
O3	4	3	6	2	5
D	6	10	15	4	35

	D1	D2	D3	D4	S
O1	6	4	1	5	14
O2	8	9	2	7	16
O3	4	3	6	2	5
D	6	10	15	4	35

$$S = 35$$

$$D = 35$$

The prob is balanced TP as
total supply = demand, we find
basic feasible soln as given
in tab 6.

Initial

	D1	D2	D3	D4	S
O1	6	4	1	5	14
O2	8	9	2	7	16
O3	4	3	6	2	5
D	6	10	15	4	35

$$1 \times 14 + 8 \times 6 + 9 \times 9 + 2 \times 1 + 3 \times 1 + 2 \times 14 = ₹ 156$$

The prob is unbalanced TP as
the total supply is greater than
total demand. $\sum b_i < \sum a_i$

Least cost Method (Unbalanced)

	D1	D2	D3	S	
O1	4	8	8	76	
O2	16	24	16	82	
O3	8	16	24	77	
D	72	102	41		

$$S = 235$$

$$D = 215$$

	D1	D2	D3	D4	S
O1	4	8	8	0	76
O2	16	24	16	0	82
O3	8	16	24	0	77
D	72	102	41	20	

	D1	D2	D3	D4	S
01	4 ⁵⁰	8	8	0 ²⁰	76
02	16 ¹⁶	24 ²⁰	16 ⁴¹	0	82
03	8	16	24	0	61
S	72 ¹⁶	102 ⁴¹	41 ⁰	20 ⁰	

$20 = 56 - 0$
 $82 = 82 - 16 = 66$
 $66 - 41 = 25$

$$4 \times 56 + 0 \times 20 + 24 \times 41 + 16 \times 41 + 8 \times 16 + 16 \times 61 = 2968$$

he can't use closed paths

VAM (balance)

$$S = 950$$

	D1	D2	D3	D4	S
01	11 ¹²⁵⁰	13 ⁵⁰	17	14	250 ²⁰
02	16	18 ¹⁷⁵	14	10 ¹²⁵	300 ²⁵
03	21	24	13 ²⁵	10 ²⁵	400 ²⁵
D	200	225	275	250	

Final cost: 175, 150, 0, 1250

$\rightarrow$ unit cost penalty
 namely the diff b/w
 smallest & next small
 cost in each row & column
 choose max penalty
 P

PS	P4	P1	P2	P3	C
-	-	2	1	1	5
-	→ 4	4	4	4	5
3	3	3	3	3	6

	D1	D2	D3	D4	S
O1	11 ¹⁰⁰	13 ¹⁰	17	14 ²⁵	250
O2	16	18 ¹²⁵	14	10	300
O3	21	24	13 ²⁷⁵	10 ¹²⁵	400
D	200	225	275	250	

No of allocated cell = $m+n-1 = 6$ & there are 6 +ve independent allocation. This ensure that the soln is nondegenerate to test C soln.

The TC

$$= 11 \times 200 + 13 \times 50 + 18 \times 175 + 10 \times 125 + 13 \times 275 + 10 \times 125$$

$$= ₹ 12075$$

VAM (Unbalanced)

	D1	D2	D3	S
O1	5	1	7	10
O2	6	4	6	80
O3	3	2	5	50
D	75	20	50	

$$D = 145$$

$$S = 140$$

≤ 6 if ≥ 6 or

	D1	D2	D3	S
O1	5	1 ¹⁰	7	100
O2	6 ¹⁰	4 ¹⁰	6 ¹⁴⁵	80 35 10
O3	3	2	5	50 0
O4	0	0	0 ⁵	5
D	75	20	50	
	25	10	45	

Find cost penalty namely the diff b/w smallest & next small cost in each row & column. Choose Max penalty

	P ₁	P ₂	P ₃	P ₄	P ₅
4	4	1	—	—	—
2	2	2	2	2	2
1	1	1	1	—	—
0	—	—	—	—	—

C

P ₁	3	1	5	1
P ₂	2	1	1	
P ₃	3	1	2	1
P ₄	6	1	4	6
P ₅	—	4	6	1

$$3 + 4 - 1 = 6$$

150

270

$$= 1 \times 10 + 6 \times 25 + 4 \times 10 + 6 \times 45 + 3 \times 50 + 0 \times 5 = ₹ 620$$

Non degenerate

	D1	D2	D3	S
O1	5	1 ¹⁰	7	10
O2	6 ²⁵	4 ¹⁰	6 ¹⁴⁵	80
O3	3 ⁵⁰	2	5	50
O4	0	0	0 ⁵	
D	75	20	50	

MODI Method (Balanced)

	D1	D2	D3	D4	S
O1	2	2	2	1	3
O2	10	8	5	4	7
O3	7	6	6	8	5
D	4	3	4	4	

Applying VAM (Method)

$$TC = ₹ 68$$

- (ii) If $\Delta_{ij} \geq 0$, then sol is opt. but alternate sol exist ^{unique} (iii) If at least one $\Delta_{ij} < 0$ then sol is not optimum (go next step)

Step 2) Applying MODI, we determine a set of no U_i & V_j for each row & column, such that $U_i + V_j = C_{ij}$ for each occupied cell.

$$\begin{aligned} C_{31} = U_3 + V_1 &\Rightarrow 7 = 0 + V_1 \Rightarrow V_1 = 7 \\ C_{32} = U_3 + V_2 &\Rightarrow 6 = 0 + V_2 \Rightarrow V_2 = 6 \\ C_{33} = U_3 + V_3 &\Rightarrow 6 = 0 + V_3 \Rightarrow V_3 = 6 \\ C_{11} = U_1 + V_1 &\Rightarrow 2 = U_1 + 7 \Rightarrow U_1 = -5 \\ C_{23} = U_2 + V_3 &\Rightarrow 5 = U_2 + 6 \Rightarrow U_2 = -1 \\ C_{24} = U_2 + V_4 &\Rightarrow 4 = -1 + V_4 \Rightarrow V_4 = 5 \end{aligned}$$

Step 3) Find the sum of U_i & V_j for each sum occupied cell then find $\Delta_{ij} = C_{ij} - (U_i + V_j)$

	D1	D2	D3	D4	U_i
O1	2	2 $\Delta=1$	2 $\Delta=1$	1 $\Delta=1$	-5
O2	10	8	5	4	-1
O3	7	6	6	8	0
V_j	7	6	6	5	

$$\begin{aligned} \Delta_{12} &= C_{12} - (U_1 + V_2) = 2 - (-5 + 6) = 3 \\ \Delta_{13} &= C_{13} - (U_1 + V_3) = 2 - (-5 + 6) = 3 \\ \Delta_{14} &= C_{14} - (U_1 + V_4) = 1 - (-5 + 5) = 1 \\ \Delta_{21} &= C_{21} - (U_2 + V_1) = 10 - (-1 + 7) = 4 \\ \Delta_{22} &= C_{22} - (U_2 + V_2) = 8 - (-1 + 6) = 3 \\ \Delta_{34} &= C_{34} - (U_3 + V_4) = 8 - (0 + 5) = 3 \end{aligned}$$

Since $\Delta_{ij} > 0$ then soln is optimum
A unique - The soln is given by,

$$x_{11} = 3, x_{23} = 3, x_{24} = 4, x_{31} = 1, x_{32} = 3, x_{33} = 1$$

MODI Method (NW)

	D1	D2	D3	D4	S
O1	3 ²⁰⁰	1 ⁰	7	4	250
O2	2 ⁺	6 ⁺	5 ⁺	9	350
O3	8	3	3 ²⁵⁰	2 ⁺	400
D					1000

$$U_1 = 0$$

$$U_2 = 5$$

$$U_3 = 3$$

$$U_1 = 3 \quad U_2 = 1 \quad U_3 = 0 \quad U_4 = -1$$

$$\text{Cost} = 3700$$

$$\Delta_{13} = 7, \Delta_{14} = 5, \Delta_{21} = -6, \Delta_{31} = 2, \Delta_{24} = 5, \Delta_{32} = -1$$

we choose $\Delta_{21} = -6$ (most negative)

$$x_{\min} = (200, 250)$$

$$x = 200$$

The Modified table as given in following table

	D1	D2	D3	D4	
O1	3	1 ²⁵⁰	7	4	$U_1 = -5$
O2	2 ²⁰⁰	6 ⁵⁰	5	9	$U_2 = 0$
O3	8	3	3	2	$U_3 = -2$

$$U_1 = 2 \quad U_2 = 6 \quad U_3 = 5 \quad U_4 = 4$$

$$\Delta_{11} = 6, \Delta_{13} = 7, \Delta_{24} = 5, \Delta_{31} = 8, \Delta_{32} = -1, \Delta_{14} = 5$$

$$\Delta_{32} = -1 \text{ \& solve } 0.5$$

$$x = 50$$

O1	3	1 ⁰	7 ⁺	4	$U_1 = -2$
O2	2 ²⁰⁰	6	5 ¹⁵⁰	9	$U_2 = 2$
O3	8	3 ⁵⁰	3 ²⁰⁰	2 ¹⁵⁰	$U_3 = 0$

$$U_1 = 0 \quad U_2 = 3 \quad U_3 = 3 \quad U_4 = 2$$

$$\Delta_{11} = 5, \Delta_{13} = 6, \Delta_{14} = 4, \Delta_{22} = 1, \Delta_{24} = 5, \Delta_{31} = 8$$

$\therefore$ optimum solution, since all $\Delta_{ij} \geq 0$

$$\text{Cost} = 1 \times 250 + 2 \times 200 + 5 \times 150 + 3 \times 50 + 3 \times 200 + 2 \times 150$$

=

Step for degeneracy

→ Among the empty cell we choose an empty cell the least cost, which is an independent position. If such cell more than one then choose any one.

② Allocate the small the quantity ϵ in that cell.

Max P

The objective is to max the total profit for which the profit matrix is given.

First we have to convert the maximization problem into min by subtracting all the element in the from the highest element in given transportation table.

Assignment Problem

Hungarian Method

Jobs	Machines				
	A	B	C	D	E
1	13	8	16	18	19
2	9	15	24	9	12
3	12	9	4	4	4
4	6	12	10	8	13
5	15	17	18	12	20

$m \times n$
 $n = 5$

Subtract the min element from all the element from all the elements in respective row, we get row reduction matrix as -

Jobs	Machine				
	A	B	C	D	E
1	5	0	8	10	11
2	0	6	15	0	3
3	8	5	0	0	0
4	0	6	4	2	7
5	3	5	0	0	8

Subtract the min element from all the element from all the element in resp col, we get col reduction matrix as

Jobs	Machine				
	A	B	C	D	E
1	5	0	8	10	11
2	0	6	15	0	3
3	8	5	0	0	0
4	0	6	4	2	7
5	3	5	0	0	8

Draw min no of horizontal & vertical line to cover all zeros.

a) If $N=n$, n =order of matrix. then an optimal soln can be made.

b) If $N < n$, then go to next step

Jobs	Machine				
	A	B	C	D	E
1	5	0	8	10	11
2	0	6	15	0	3
3	8	5	0	0	2
4	0	6	4	2	7
5	3	5	6	0	8

Row 1 zero-Marked
Row 2 Max zero skip

Determine the smallest uncovered element (x)

a) Write uncovered value = uncovered value - x

b) Intersection value = intersection value + x

c) Line value (other values) as same.

$$x = 3$$

Jobs	Machine				
	A	B	C	D	E
1	2	0	5	7	8
2	0	3	12	0	0
3	5	2	0	3	0
4	0	3	1	0	4
5	0	2	3	0	5

Jobs	Machine				
	A	B	C	D	E
1	2	0	5	7	8
2	0	3	12	0	0
3	5	2	0	3	0
4	0	3	1	0	4
5	0	2	3	0	5

Go to step 3

$$n = 5, N = 5$$

Job	Machine
1	B
2	E
3	C
4	A
5	D

Min(total cost) =

$$8 + 12 + 4 + 6 + 12 = 42$$

	J_1	J_2	J_3	J_4
M_1	10	9	8	7
M_2	8	4	5	6
M_3	2	1	1	2
M_4	4	3	5	6

$$n=4$$

~~max~~
4x4

~~12=0~~

R

C

	J_1	J_2	J_3	J_4
M_1	3	2	1	0
M_2	0	1	2	3
M_3	1	0	0	1
M_4	1	0	2	3

	J_1	J_2	J_3	J_4
M_1	3	2	1	0
M_2	0	1	2	3
M_3	1	0	0	1
M_4	1	0	2	3

R → one zero X
two zero —

	J_1	J_2	J_3	J_4
M_1	3	2	1	0
M_2	0	1	2	3
M_3	1	0	0	1
M_4	1	0	2	3

$$n=N$$

$$x=1$$

$$7 + 3 + 1 + 3 = 14$$

$$x_1 + x_2 = 30$$

	J_1	J_2	J_3	J_4
M_1	3	2	1	0
M_2	0	1	2	3
M_3	1	0	0	1
M_4	1	0	2	3

- x_1 = trained machine in Jan
 x_2 = idle machine in Jan
 x_3 = trained machine in Feb
 x_4 = idle machine in Feb
 x_5 = trained machine in Mar
 x_6 = idle machine in Mar

Jan
 $100 + x_1 + x_2 = 130$

Feb
 $150 + x_2 + x_3 = 130 + 7x_1$

Mar
 $200 + x_4 + x_5 = 130 + 7x_1 + 7x_3$

$$130 + 7x_1 + 7x_3 + 7x_5 = 250$$

$$400(10x_1 + 10x_3 + 10x_5) + 700(x_1 + x_3 + x_5) + 500(x_2 + x_4 + x_6)$$

8)

	J_1	J_2	J_3	J_4
M_1	10	9	7	8
M_2	5	8	7	7
M_3	5	4	6	5
M_4	2	3	4	5

$$N=4$$

Row $\pi=2$

	J_1	J_2	J_3	J_4
M_1	3	2	0	1
M_2	0	3	2	2
M_3	1	0	2	1
M_4	1	1	2	3

column

	J_1	J_2	J_3	J_4
M_1	3	2	0	0
M_2	0	3	2	1
M_3	1	0	2	0
M_4	1	1	2	2

x_1	0	10
x_2	5	0

$$x_1 + 2x_2 \leq 10$$

x_1	0	6
x_2	6	0

$$x_1 + x_2 \leq 6$$

x_1	0	2
x_2	2	0

$$x_1 - x_2 \leq 2$$

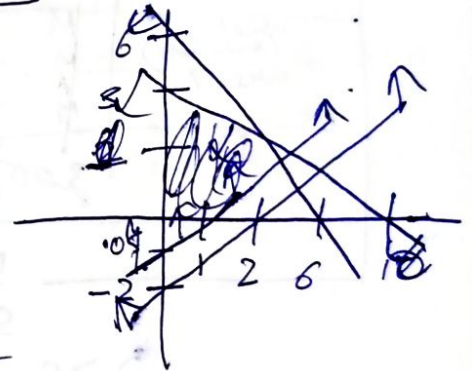
x_1	0	1
x_2	1	0

$$x_1 - 2x_2 \leq 1$$

	J_1	J_2	J_3	J_4
M_1	3	2	0	1
M_2	0	3	2	2
M_3	1	0	2	1
M_4	1	1	2	3

 $\pi=1$

	J_1	J_2	J_3	J_4
M_1	4	2	0	0
M_2	0	2	1	0
M_3	2	0	2	0
M_4	1	0	1	1

 $n=4$ 

Optimal soln

M_1	J_3
M_2	J_1
M_3	J_4
M_4	J_2

Total cost

$$7 + 5 + 5 + 3 = 20$$

$$40000x_1 + 75000x_2 + 30000x_3 + 15000x_4 \leq 80000$$

$$300000x_1 + 400000x_2 + 200000x_3 + 100000x_4 \geq 2000000$$

$$400000x_1 + 900000x_2 + 500000x_3 + 200000x_4 \leq 500000$$

$$x_1 \geq 3 \quad 5 \leq x_3 \leq 10$$

$$x_2 \geq 2 \quad 5 \leq x_4 \leq 10$$

Smith
Jones
Richard
Benny
D

A	B	S
13	17.5	2000
15	30.5	30050
12	31	4000
0	0	10000
450	550	
250	500	

$$D = 1000$$

$$S = 900$$

$$D > S$$

$$13 \times 200 + 15 \times 250 + 30.5(50) + 31 \times 400 + 0 \times 100$$

$$2600 + 3750 + 1525 + 12400 + 0$$

$$= 20275$$

$$50 - (10 + 2 + 25) = 13$$

$$50 - (9 + 1 + 25) = 15$$

$$50 - (8 + 5 + 25) = 12$$

$$50 - (10 + 2.5 + 20) = 17.5$$

$$50 - (9 + 1.5 + 20) = 30.5$$

$$50 - (8 + 3 + 20) = 31$$

3.4

	week 1	week 2	week 3	week 4	DW	S
week 1	10	13	16	19	0	200 4000
week 2	M	10	13	16	0	200 400
week 2 (overtime)	M	15	18	21	0	200
week 3 (overtime)	M	M	15	18	0	200
week 4	M	M	20	23	0	200
D	M	M	M	15	0	700
	3000	200 300	900 300	800 100		

$$D = 2700$$

$$S = 2500$$

$$D > S$$

3.2

	M ₁	M ₂	M ₃	M ₄	S
W ₁	2	2	2	13	3
W ₂	10	8	5	4	7
W ₃	7	6	6	8	5
D	4	3	4	4	

1x3
8x2
5x4
4x1
7x4
6x1

R

P ₁	P ₂	P ₃	P ₄	P ₅
1	-	-	-	-
1	1	3	3	8
1	1	1	0	-

C

P ₁	2	1	2	3
P ₂	3	2	1	4
P ₃	3	2	1	-
P ₄	-	2	1	-
P ₅	-	2	-	-

fly

	M1	M2	M3	M4	S
W1	2	2	2	1	3
W2	10	8	5	4	7
W3	7	6	6	8	5
D	4	3	4	4	

$$2 \times 3 + 5 \times 3 + 4 \times 4 + 7 \times 1 + 6 \times 3 + 6 \times 1 = 68$$



R	P1	P2	P3	P4	P5
1	1	3	1	3	5
2	1	1	0	0	4

R	6	4	1	4
P2	3	2	1	4
P3	3	2	1	4
P4	-	2	1	-
P5	-	-	1	-

$$3x_1 + x_2 +$$

Q1

$$Z = 4x_1 + 2x_2 + 3x_3$$

sub to constraints

$$7x_1 + 3x_2 + 6x_3 \leq 150$$

$$4x_1 + 4x_2 + 5x_3 \leq 200$$

~~4x1~~

$$x_1, x_2, x_3 \geq 0$$

$$Z = 2x_1 + 3x_2 + 4x_3$$

sub to const

$$3x_1 + x_2 + 4x_3 \leq 600$$

$$2x_1 + 4x_2 + 2x_3 \leq 480$$

$$2x_1 + 3x_2 + 3x_3 = 540$$

Q2) ~~Z = 400~~

Blank, Surplus & Artificial variables

Infeasible, unbounded & degenerate LP problem

degeneracy in LPP

~~Feasibility analysis in LPP~~

~~Method to find Basic Feasible soln in LP~~

Duality in LPP

Best (opt)

Artificial variable are those variable which is a fictitious & can not have any physical meaning. It is merely a device to get the starting basic feasible soln, so that the simplex ~~pro~~ procedure may be adopted as usual until the optimal soln is obtained. Cost of artificial variable is M.

Slack variable are those variable that is added to an inequality constraint to transform it into an equality. Introducing a slack variable replaces an inequality constraint with an equality constraint & a non-negativity constraint on slack variable.

Surplus variable is also known as negative slack variable. It is a variable that is subtracted to an inequality constraint to transform it into an equality.

Infeasible soln are those soln in which no combination of decision variable values can satisfy the entire set of requirement & constraints.

Unbounded soln are those soln in LPP is a situation where objective fn is infinite. A LPP is said to ~~be~~ have unbounded soln if its soln can be made infinitely large without violating any constraints.

Degeneracy in LPP is said to occur when a basic feasible solution contains a smaller no of non-zero variables than the no of independent constraints.

Sensitivity analysis is an analysis of the effect of optimal soln of changes in the parameter of a problem on current optimal soln.

Sensitivity analysis allows for forecasting using historical, true data. By studying all the variables & possible outcome. It is of 4 types

- ① Change in objective fn
- ② Change in right ^{hand} ~~side~~ ranging
- ③ Change in tech coefficient
- ④ Adding or ~~deleting~~ constraints

Duality states that every LPP has another LPP related to it & thus can be derived from it. The original LPP is called 'Primal' while derived LPP is called 'Dual'.

PERT:- (Program Evaluation Review technique). It is a tool used in project Management. It is used to identify task dependencies, plan resources, estimate task duration.

CPM (Critical path Method). It is a technique where we identify tasks that are necessary for project completion & determine scheduling flexibility.

Pert

- i) Stands for programme evaluation & review technique
- ii) Event oriented
- iii) Associated with probabilistic activity
- iv) Based on 3 times estimate
- v) Resource such as labour, equipments material are limited
- vi) Mainly used for research & development project
- vii) Planning & control of time
- viii) The nature of the job is non-repetitive

~~CPM~~ **CPM**

i) Stands for critical path Method.

- (ii) Activity oriented.
- (iii) Associated with deterministic activity.
- (iv) Based on single time to complete.
- v) No limitation of resources.
- vi) Mainly used for construction project.
- vii) Control cost & time
- viii) It involves the job of repetitive nature.

$$\text{Min } Z = 7x_1 + 15x_2 + 20x_3$$

$$\text{Sub to } 2x_1 + 4x_2 + 6x_3 \geq 24$$

$$3x_1 + 9x_2 + 6x_3 \geq 30$$

Soln

$$7x_1 + 15x_2 + 20x_3 + 0x_4 + 0x_5 + Mx_6 + Mx_7$$

$$2x_1 + 4x_2 + 6x_3 - x_4 + 0x_5 + x_6 + 0x_7 = 24$$

$$3x_1 + 9x_2 + 6x_3 + 0x_4 - x_5 + 0x_6 + x_7 = 30$$

		C_j	7	15	20	0	0	M	M	$x_1, x_2, x_3, x_4, x_5, x_6, x_7 \geq 0$	
		b	x_1	x_2	x_3	x_4	x_5	x_6	x_7	Min	Ratio
x_6	M	24	2	4	6	-1	0	1	0		
x_7	M	30	3	9	6	0	-1	0	1		

C_B	x_B	C_j	b	x_1	x_2	x_3	x_4	x_5	x_6	x_7	Min Ratio
M	x_6	24	2	4	6	-1	0	1	0	0	24
M	x_7	30	3	9	6	0	-1	0	1	1	

Z_j 5M 13M 12M -M -M M M 3 24
 $C_j - Z_j$ 7-5M 15-13M 20-12M M M +0 0

C_B	x_B	C_j	b	x_1	x_2	x_3	x_4	x_5	x_6	x_7	Min Ratio
M	x_6	24	2	4	6	-1	0	1	0	0	24/4 = 6
M	x_7	30	3	9	6	0	-1	0	1	1	30/9 = 3.3 ← leaving

Z_j 5M 13M 12M -M -M M M
 $C_j - Z_j$ 7-5M 15-13M 20-12M M M

C_B	x_B	C_j	b	x_1	x_2	x_3	x_4	x_5	x_6	Min Ratio
M	x_6	32/3	2/3	0	5/3	-11/3	4/9	1	0	32/3 × 1/2 = 16/3
15	x_2	30/9	1/3	1	2/3	0	-1/9	0	0	30/9 × 2/3 = 20/3

Z_j $\frac{2M+15}{3}$ 15 $\frac{5M+30}{3}$ $\frac{-11M}{3}$ $\frac{4M-15}{9}$ M
 $C_j - Z_j$ $\frac{6+2M}{3}$ 0 $\frac{10-2M}{3}$ $\frac{11M}{3}$ $\frac{-4M+15}{9}$ 0

C_B	x_B	C_j	b	x_1	x_2	x_3	x_4	x_5	x_6	Min Ratio
0M	x_6	32/3	2/3	0	10/3	-1	4/9	1	0	32/3 × 3/10 = 32/10
15	x_2	30/9	1/3	1	2/3	0	-1/9	0	0	30/9 × 3/2 = 20/3

Z_j $\frac{2M+15}{3}$ 15 $\frac{10M+10}{3}$ -M $\frac{4M-5}{9}$ M
 $C_j - Z_j$ $\frac{6+2M}{3}$ 0 $\frac{-10M+30}{3}$ M $\frac{-4M+5}{9}$ 0

C_B	x_B	C_j	b	x_1	x_2	x_3	x_4	x_5
20	x_3	16/15	1/15	0	1	-3/10	2/15	0
15	x_2	6/5	1/5	1	0	1/5	-1/5	0

Z_j 20 15
 $C_j - Z_j$ 0 0

$$1 - 4\left(\frac{2}{3}\right) = 1 - \frac{8}{3} = \frac{3-8}{3} = -\frac{5}{3}$$

$$-\frac{5}{3}M + \frac{30}{3} = \frac{30-8M}{3}$$

$$-\frac{5M}{3} + \frac{30}{3}$$

$$\frac{32 \times 3}{3} = \frac{96}{3}$$

$$\begin{array}{r} 3 \\ 24 \\ \times 9 \\ \hline 216 \end{array}$$

$$\begin{array}{r} 111 \\ 216 \\ \times 6 \\ \hline 120 \end{array}$$

$$24 - 4\left(\frac{30}{9}\right) = 24 - \frac{120}{9} = \frac{216-120}{9}$$

$$x_1 = 0$$

$$x_2 = 6/5$$

$$x_3 = 16/5$$

$$0L - \frac{2}{3}NL$$

$$30/9 - \frac{2}{3}\left(\frac{32}{9}\right)$$

$$1/3 - \frac{2}{3}\left(\frac{2}{10}\right)$$

$$1 - \frac{2}{3}\left(\frac{0}{10}\right)$$

$$2/3 - 1\left(\frac{2}{3}\right) = 0$$

$$0 - 2/3(-3/10)$$

$$-1/9 - 2/3(2/15)$$

$$\frac{30}{9} - \frac{64}{30} = \frac{\quad}{30}$$

$$7(0) + 15\left(\frac{6}{5}\right) + 20\left(\frac{16}{5}\right)$$

Max

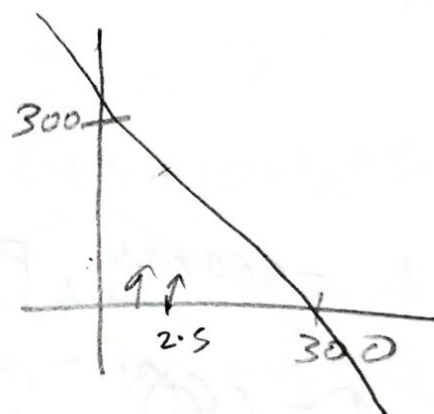
$$Z = 10x_1 + 8x_2$$

$$2000x_1 + 1000x_2$$

x_1	0	300
x_2	300	0

$$x_1 + x_2 \geq 300 \quad \text{--- (1)}$$

$$2000x_1 + 1000x_2 \geq 500$$



x_1	0	5/2
x_2	1/2	0

$$2x_1 + 10x_2 \geq 5 \quad \text{--- (2)}$$

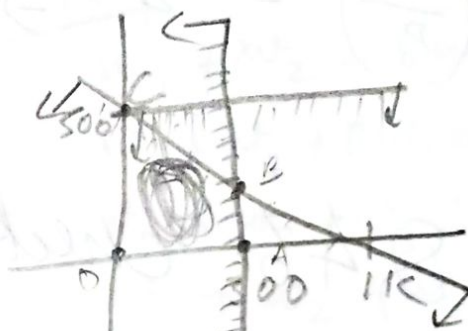
$$x_1, x_2 \geq 0$$

x_1	0	1000
x_2	500	0

$$2x_1 + x_2 \leq 1000$$

$$x_1 \leq 300$$

$$x_2 \leq 500$$



$$O(0,0) = 0$$

$$A(300,0) = 3000$$

$$B(300,100) = 6200$$

$$C(0,500) = 4000$$

$$500 \times 8 = 4000$$

$$2x_1 + x_2 = 1000$$

$$x_1 = 300$$

$$x_2 = 400$$

$$2 \times 300 + 400 = 1000$$

Z =

$$2(300) + 400 = 1000 + 400 = 1400$$

$$3000 + 3000 + 3200$$

$$6200$$

Max

Q1)

Min

~~25w₁~~

$$25w_1 + 15w_2 + 0w_3$$

$$w_1 + 2w_2 \geq 1$$

$$2w_1 + w_2 \geq 2$$

$$2w_1 - 3w_2 \geq -3$$

$$-3w_1 + 2w_2 \geq 4$$

Q3

$$2w_1 + w_2$$

Min

$$-w_1 - 2w_2 \geq 1$$

$$w_1 + w_2 \geq 1$$

$$w_1 + w_2 \geq 0$$

Q5

$$2w_1 + w_2 + 4w_3$$

Min

$$2w_1 + 3w_2 + w_3 \geq 1$$

$$3w_1 - w_2 + w_3 \geq 4$$

$$-5w_1 + 6w_2 + w_3 = 3$$

Q4

$$z = 4x_1 + 5x_2$$

$$-3x_1 + 2x_2 \leq 20$$

$$4x_1 - 3x_2 \geq 10$$

$$x_1 + x_2 = 5$$

$$x_1 \geq 0, x_2 = \text{unrest}$$

$$z = 4x_1 + 5x_2 - 5x_4$$

$$-3x_1 + 2x_2$$

Q2

$$\text{Min } 20w_1 + 20w_2$$

$$w_1 + 2w_2 \geq 1$$

$$2w_1 + w_2 \geq 2$$

$$2w_1 + 3w_2 \geq 3$$

$$3w_1 + 2w_2 \geq 4$$

~~Q4~~

$$20w_1 + 10w_2 + 5w_3$$

Min

$$3w_1 + 4w_2 + w_3 \geq 4$$

$$2w_1 - 3w_2 + w_3 \geq 5$$

$$0w_1 + 0w_2 + 0w_3 \geq 0$$

Q6

$$w_1 + 2w_2 + 3w_3$$

$$w_1 + w_2 + 0w_3 = 2$$

$$w_1 - w_2 + w_3 \leq 1$$

$$-w_1 + w_2 + w_3 \geq -1$$

$$20w_1 + 10w_2$$

$$4x_1 + 5x_2$$

$$-3x_1 + 2x_2 \geq -20$$

$$4x_1 - 3x_2 \geq 10$$

$$x_1$$

$$x_2 = (x_3 - x_4)$$

$$\text{Min } Z = 7x_1 + 15x_2 + 20x_3$$

Sub to,

$$2x_1 + 4x_2 + 6x_3 \geq 24$$

$$3x_1 + 9x_2 + 6x_3 \geq 30$$

$$x_1, x_2, x_3 \geq 0$$

$$30 - 24 =$$

$$Z = 7x_1 + 15x_2 + 20x_3 + 0x_4 + 0x_5$$

$$2x_1 + 4x_2 + 6x_3 + x_4 + 0x_5 \geq 24$$

$$3x_1 + 9x_2 + 6x_3 + 0x_4 + x_5 = 30$$

$$x_1, x_2, x_3, x_4, x_5 \geq 0$$

$$3 - \frac{6}{3} = \frac{3}{3}$$

OL-NL

$$30 - 6(4) = 6$$

$$3 - 6(1/3) = 1$$

$$9 - 6(2/3) = 5$$

$$6 - 6(1) = 0$$

$$0 - 6(1/6) = -1$$

$$1 - 6(0) = 1$$

$$\frac{15}{24} = \frac{5}{8}$$

$$\frac{5}{8} = \frac{5}{8}$$

OL-NL

$$24 - 4(10/3) = 2/3$$

$$2 - 4(1/3) = 2/3$$

$$4 - 4(1) = 0$$

$$6 - 4(2/3) = 10/3$$

$$1 - 4(0) = 1$$

$$0 - 4(1/9) = 4/9$$

$$4 - \frac{2}{3}(\frac{6}{5}) = \frac{10}{3}$$

$$4 - \frac{12}{15} = \frac{16}{15}$$

$$60 - 12 = 48$$

$$\frac{48}{15} = \frac{16}{5}$$

OL-NL

$$4 - 2/3(6/5) = 16/5$$

$$1/3 - 2/3(1) = -1/3$$

$$2/3 - 2/3(1) = 0$$

$$1 - 2/3(0) = 1$$

$$1/6 - 2/3(1/5) = 10/3 - 2/3(16/5)$$

$$0 - 2/3(1/5) = 1/3 - 2/3(1/5)$$

$$-2/3 - 2/3(1) = -2/3 - 2/3(1)$$

$$1/6 - 2/3(3/10) = 1/6 - 2/3(2/15)$$

$$1/6 - 2/3(1/5) = 1/6 - 2/3(1/5)$$

$$1/6 - 2/3(1/5) = 1/6 - 2/3(1/5)$$

$$1/6 - 2/3(1/5) = 1/6 - 2/3(1/5)$$

$$1/6 - 2/3(1/5) = 1/6 - 2/3(1/5)$$

$$1/6 - 2/3(1/5) = 1/6 - 2/3(1/5)$$

C_b	x_b	C_j	7	15	20	0	0	
		b	x_1	x_2	x_3	x_4	x_5	Min R
0	x_4	24	2	4	6	1	0	$24/4 = 6$
0	x_5	30	3	9	6	0	1	$30/9 = 3\frac{1}{3}$ leaving

$$Z_j \quad 0 \quad 0 \quad 0 \quad 0 \quad 0$$

$$C_j - Z_j \quad 7 \quad 15 \quad 20 \quad 0 \quad 0$$

C_b	x_b	C_j	7	15	20	0	0	
		b	x_1	x_2	x_3	x_4	x_5	Min R
0	x_4	$32/3$	$2/3$	0	$10/3$	1	$4/9$	$32/10 = 3\frac{1}{5}$ leaving
15	x_2	$10/3$	$1/3$	1	$2/3$	0	$1/9$	$10/2 = 5$

$$Z_j \quad 5 \quad 15 \quad 10 \quad 0 \quad 5/3$$

$$C_j - Z_j \quad 2 \quad 0 \quad 10 \quad 0 \quad -5/3$$

C_b	x_b	C_j	7	15	20	0	0	
		b	x_1	x_2	x_3	x_4	x_5	Min R
20	x_3	$16/5$	$1/5$	0	1	$3/10$	$2/15$	
15	x_2	$6/5$	$1/5$	1	0	$-2/15$	$1/45$	

$$Z_j \quad 1 \quad 15 \quad 20 \quad 4 \quad 10/5$$

$$C_j - Z_j \quad 6 \quad 0 \quad 0 \quad -4 \quad -10/5$$

$$x_1 = 0$$

$$x_2 = 6/5$$

$$x_3 = 16/5$$

$$\frac{1}{6} - \frac{2}{3}(\frac{1}{5}) = \frac{1}{6} - \frac{2}{15} = \frac{1}{10}$$

$$\frac{15 - 12}{15} = \frac{3}{15} = \frac{1}{5}$$

$$\therefore Z = 7(0) + 15\left(\frac{6}{5}\right) + 20\left(\frac{16}{5}\right)$$

$$= \frac{90}{5} + \frac{320}{5} = 82$$

$$\boxed{Z = 82}$$

$$\text{Min } Z = -3x_1 + x_2 + x_3$$

sub to,

$$x_1 - 2x_2 + x_3 \leq 11$$

$$-4x_1 + x_2 + 2x_3 \geq 3$$

$$2x_1 - x_3 = -1$$

$$Z = -3x_1 + x_2 + x_3 + 0x_4 + 0x_5 + Mx_6 + Mx_7$$

sub to,

$$x_1 - 2x_2 + x_3 + x_4 + 0x_5 + 0x_6 + 0x_7 = 11$$

$$-4x_1 + x_2 + 2x_3 + 0x_4 - x_5 + 0x_6 + 0x_7 = 3$$

$$-2x_1 + x_3 + 0x_4 + 0x_5 + 0x_6 + x_7 = 1$$

C_b	x_b	C_j	-3	1	1	0	0	M	M	
		b	x_1	x_2	x_3	x_4	x_5	x_6	x_7	Min R
0	x_4	11	1	-2	1	1	0	0	0	11/1=11
M	x_6	3	-4	1	2	0	-1	1	0	3/2=1.5
M	x_7	1	-2	0	1	0	0	0	1	1/1=1

$$Z_j \quad -6M \quad 3M \quad 3M \quad 0 \quad -M \quad M \quad M$$

$$C_j - Z_j \quad -3-6M \quad -3-M \quad 1-3M \quad 0 \quad M \quad 0 \quad 0$$

C_b	x_b	C_j	-3	1	1	0	0	M	
		b	x_1	x_2	x_3	x_4	x_5	x_6	Min R
0	x_4	10	3	-2	0	1	0	0	10/3
M	x_6	1	0	1	0	0	-1	1	1/1=1
1	x_3	1	-2	0	1	0	0	0	1/0

$$Z_j \quad -2 \quad M \quad 1 \quad 0 \quad -M \quad M$$

$$C_j - Z_j \quad -1 \quad 1-M \quad 0 \quad 0 \quad M \quad 0$$

$$OL-NL$$

$$11-1(1)=10$$

$$1-1(2)=3$$

$$-2-1(0)=-2$$

$$1-1(1)=0$$

$$1-1(0)=1$$

$$0-1(0)=0$$

$$0-1(0)=0$$

$$OL-NL$$

$$3-2(0)=3$$

$$-4-2(-2)=0$$

$$1-2(0)=1$$

$$2-2(1)=0$$

$$0-2(0)=0$$

$$-1-2(0)=-1$$

$$1-2(0)=1$$

$$-\frac{3}{3} + \frac{2}{3}$$

$$OL-NL$$

$$10+2(1)=12$$

$$3+2(0)=3$$

$$-2+2(1)=0$$

$$0+2(0)=0$$

$$1+2(0)=1$$

$$0+2(-1)=-2$$

$$OL-NL$$

$$1-1(1)=0$$

$$-2-1(0)=-2$$

$$0-1(1)=-1$$

$$1-1(0)=0$$

$$0-1(0)=0$$

$$0-1(-1)=1$$

$$OL-NL$$

$$1+2(4)=9$$

$$-2+2(1)=0$$

$$0+2(0)=0$$

$$1+2(0)=1$$

$$0+2(1/3)=2/3$$

C_b	x_b	b	-3 x_1	1 x_2	1 x_3	0 x_4	0 x_5	Min R
0	x_1	12	3	0	0	1	-2	$12/3 = 4$ leaving
1	x_2	1	0	0	0	0	-1	$1/0 = \infty$
1	x_3	1	-2	1	1	0	0	$1/-2 = \infty$
	Z_j		-2	1	1	0	-1	

C_b	x_b	4	-3	1	1	0
		b	x_1	x_2	x_3	x_4
-3	x_1	4	1	0	0	1/3
1	x_2	1	0	1	0	0
1	x_3	9	0	0	1	2/3
	Z_j	-3	1	1		1/3

$$x_1 = 4, x_2 = 1, x_3 = 9$$

$$= -12 + 1 + 9 = -2$$

	M ₁	M ₂	M ₃	M ₄	S
w ₁	2 ³	2	2	1	3
w ₂	10	8	5 ³	4 ⁴	7-4=3
w ₃	7 ¹¹	6 ³	6 ¹¹	8	5-1=4 4-3=1

R

P₁

P₂

P₃

P₄

P₅

1	1	1	3	③↑	-
1	1	1	1	0	0

cost

	J_1	J_2	J_3	J_4
M_1	10	9	7	8
M_2	5	8	7	7
M_3	5	4	6	5
M_4	2	3	4	5

$$n = 4 \times 4$$

Row wise (smallest)

	J_1	J_2	J_3	J_4
M_1	3	2	0	1
M_2	0	3	2	2
M_3	1	0	2	1
M_4	0	1	2	3

Column wise (smallest)

	J_1	J_2	J_3	J_4
M_1	3	2	0	0
M_2	0	3	2	1
M_3	1	0	2	0
M_4	0	1	2	2

$$K=1$$

	J_1	J_2	J_3	J_4
M_1	3	2	0	1
M_2	0	3	2	2
M_3	1	0	2	1
M_4	0	1	2	3

	J_1	J_2	J_3	J_4
M_1	3	2	0	0
M_2	0	3	2	1
M_3	1	0	2	0
M_4	0	1	2	2

	J_1	J_2	J_3	J_4
M_1	3	2	0	0
M_2	0	3	2	1
M_3	1	0	2	0
M_4	0	1	2	2

	J_1	J_2	J_3	J_4
M_1	3	2	0	0
M_2	0	3	2	1
M_3	1	0	2	0
M_4	0	1	2	2

	J_1	J_2	J_3	J_4
M_1	4	2	0	0
M_2	0	2	1	0
M_3	2	0	2	0
M_4	0	0	1	1

$M_1 \rightarrow J_3$
 $M_2 \rightarrow J_1$
 $M_3 \rightarrow J_4$
 $M_4 \rightarrow J_2$

$$\text{Min (total cost)} = 7 + 5 + 5 + 3 = 20$$

$$7 - \frac{20}{3} = \frac{21-20}{3} = \frac{1}{3}$$

$$15 - \frac{40}{3} = \frac{45-40}{3} = \frac{5}{3}$$

$$\text{Min } z = 7x_1 + 15x_2 + 20x_3$$

Sub to

$$2x_1 + 4x_2 + 6x_3 \geq 24$$

$$3x_1 + 9x_2 + 6x_3 \geq 30$$

$$x_1, x_2, x_3 \geq 0$$

$$z = 7x_1 + 15x_2 + 20x_3 + 0x_4 + 0x_5$$

$$2x_1 + 4x_2 + 6x_3 + x_4 + 0x_5 = 24$$

$$3x_1 + 9x_2 + 6x_3 + 0x_4 + x_5 = 30$$

$$x_1, x_2, x_3, x_4, x_5 \geq 0$$

C_b	x_b	C_i	x_1	x_2	x_3	x_4	x_5	Min R
0	x_4	24	2	4	<u>6</u>	1	0	$24/6 = 4 \leftarrow \text{leaving}$
0	x_5	30	3	9	6	0	1	$30/6 = 5$

$$z_j \quad 0 \quad 0 \quad 0 \quad 0 \quad 0$$

$$C_i - z_j \quad 7 \quad 15 \quad 20 \quad 0 \quad 0$$

↑
entering

C_b	x_b	C_i	x_1	x_2	x_3	x_4	x_5	Min R
20	x_3	4	$1/3$	$2/3$	1	$1/6$	0	$2 \cdot 3/8$
0	x_5	6	1	<u>5</u>	0	1	1	$1 \cdot 6/5 \leftarrow \text{leaving}$

$$z_j \quad 20/3 \quad 40/3 \quad 20 \quad 10/3 \quad 0$$

$$C_i - z_j \quad 1/3 \quad 5/3 \quad 0 \quad -10/3 \quad 0$$

↑
entering

C_b	x_b	C_i	x_1	x_2	x_3	x_4	x_5	
20	x_3	$16/5$	$-1/3$	0	1	$1/5$	$-2/15$	
15	x_2	$6/5$	1	1	0	$1/5$	$1/5$	

$$x_1 = 0$$

$$x_2 = 6/5$$

$$x_3 = 16/5$$

$$7(0) + 15(6/5) + 20(16/5)$$

$$90 + \frac{320}{5} = 82$$

$$z = 82$$

$$\text{Min } Z = 7x_1 + 15x_2 + 20x_3$$

sub to

$$2x_1 + 4x_2 + 6x_3 \geq 24$$

$$3x_1 + 9x_2 + 6x_3 \geq 30$$

$$x_1, x_2, x_3 \geq 0$$

$$Z = 7x_1 + 15x_2 + 20x_3 + 0x_4 + 0x_5 + Mx_6 + Mx_7$$

$$2x_1 + 4x_2 + 6x_3 - x_4 + 0x_5 + x_6 + 0x_7 = 24$$

$$3x_1 + 9x_2 + 6x_3 + 0x_4 - x_5 + 0x_6 + x_7 = 30$$

$$\begin{array}{r} 15 - 13M \\ 15 - 13M \\ 20 - 1 \end{array}$$

C_b	x_b	b	x_1	x_2	x_3	x_4	x_5	x_6	x_7	Min R
M	x_6	24	2	4	6	-1	0	1	0	$24/4 = 6$
M	x_7	30	3	9	6	0	-1	0	1	$30/9 \leftarrow$

4 - 4(1)

leaving

$$\begin{array}{l} Z_j \quad 5M \quad 13M \quad 12M \quad -M \quad -M \quad M \quad M \\ C_j - Z_j \quad 7-5M \quad 15-13M \quad 20-12M \end{array}$$

$$\begin{array}{l} 0L - NL \\ 0 - 4(1/9) \\ 1 - 4(0) \end{array}$$

C_b	x_b	b	x_1	x_2	x_3	x_4	x_5	x_6	Min R
M	x_6	$32/3$	$2/3$	0	$10/3$	1	$-4/9$	1	$32/3 \times 3/10 = 32/10 \leftarrow$
15	x_2	$30/9$	$1/3$	1	$2/3$	0	$-1/9$	0	$30/9 \times 3/2 = 5$

$$Z_j \quad \frac{2M+15}{3} \quad 15 \quad \frac{10M+30}{3} \quad -M \quad \frac{4M-5}{9} \quad M$$

$$C_j - Z_j \quad \frac{2M+6}{3} \quad 0 \quad \frac{-10+30}{3} \quad M \quad \frac{-4M+5}{9} \quad 0$$

$$\begin{array}{r} 32 \times 3 \\ 3 \times 10 \\ 10 \\ 30 \times 3 \\ 9 \times 2 \end{array}$$

C_b	x_b	b	x_1	x_2	x_3	x_4	x_5
20	x_3	$16/5$	$1/5$	0	1	$3/10$	$2/15$
15	x_2	$6/5$	$1/5$	1	0	$-2/15$	$1/45$

$$\begin{array}{l} x_1 = 0 \\ x_2 = 6/5 \\ x_3 = 16/5 \end{array}$$